Speech Understanding

Multi-Stage Cepstral Feature Extraction & Phoneme Boundary Detection

1. Introduction

This assignment implemented a manual cepstral speech-processing pipeline based on lecture concepts from cepstrum and windowing. The work covered four parts: manual MFCC/cepstrum extraction, spectral leakage and SNR analysis for different windows, voiced/unvoiced boundary detection, and phonetic mapping using a pretrained Transformer model. The goal was to understand the full signal-processing pipeline without relying on high-level functions such as librosa.feature.mfcc.

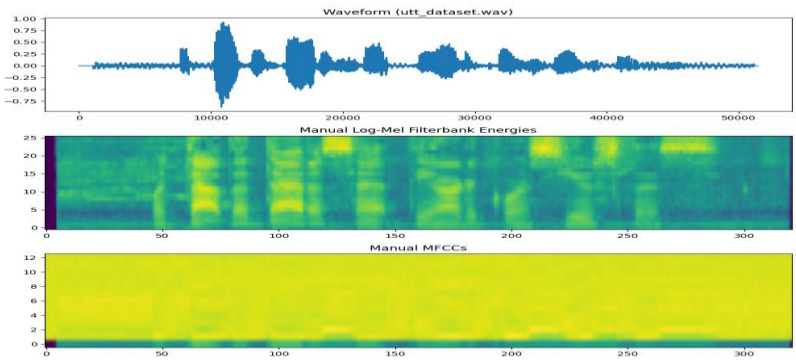
2. Manual MFCC and Cepstrum Extraction

A handcrafted MFCC engine was implemented using the standard sequence of operations: pre-emphasis, framing, windowing, FFT, Mel filterbank, log compression, and DCT. The main hyperparameters were:

- Sampling rate: 16 kHz
- Pre-emphasis coefficient: 0.97
- Frame length: 25 ms
- Frame shift: 10 ms
- FFT size: 512
- Window: Hamming
- Number of Mel filters: 26
- Number of cepstral coefficients: 13

The extracted log-Mel features captured the spectral envelope, while the MFCCs provided a compact cepstral representation. The cepstrum was also useful for separating low-quefrency envelope information and high-quefrency pitch-related information, which was used later for boundary detection.

Representative figure:



3. Spectral Leakage and SNR Analysis

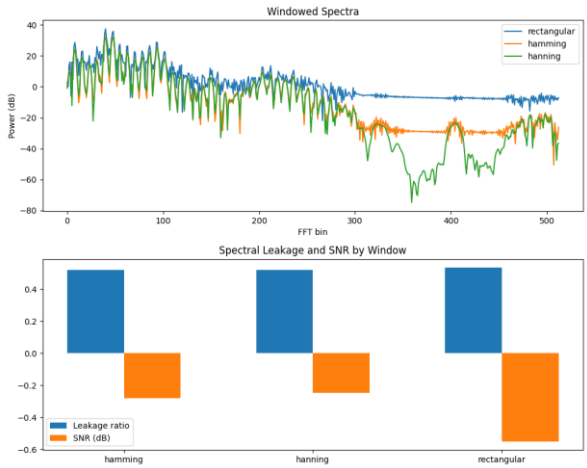
Spectral leakage and SNR were compared for three window functions: Rectangular, Hamming, and Hanning. A short speech segment was analyzed under each window, and the results are shown below.

Table 1. Spectral Leakage and SNR Comparison

Window	Spectral Leakage Ratio	SNR (dB)
Rectangular	0.5334	-0.5505
Hamming	0.5203	-0.2801
Hanning	0.5185	-0.2479

In this selected speech segment, the rectangular window produced the highest leakage ratio and the lowest SNR. Since the experiment used a real speech signal rather than an ideal sinusoid, the outcome is signal-dependent and should not be overgeneralized. The comparison still shows how strongly the choice of window affects spectral analysis.

Representative figure:

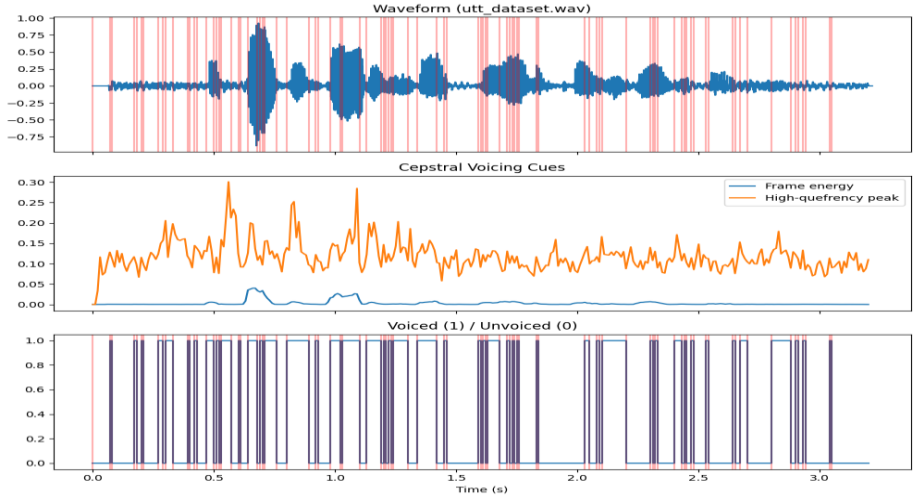


4. Voiced/Unvoiced Boundary Detection

Voiced and unvoiced sections were detected using cepstral cues. The low-quefrency region was used as a measure of the vocal-tract envelope, while the high-quefrency region was used to capture pitch-related periodicity. Frame energy, low-quefrency magnitude, and high-quefrency peaks were combined to classify each frame as voiced or unvoiced.

This produced an automatic segmentation of the speech signal into broad voiced and unvoiced regions. Although it is not a full phoneme boundary detector, it provides meaningful linguistic structure and shows how cepstrum can be used beyond feature extraction.

Representative figure



5. Phonetic Mapping and RMSE

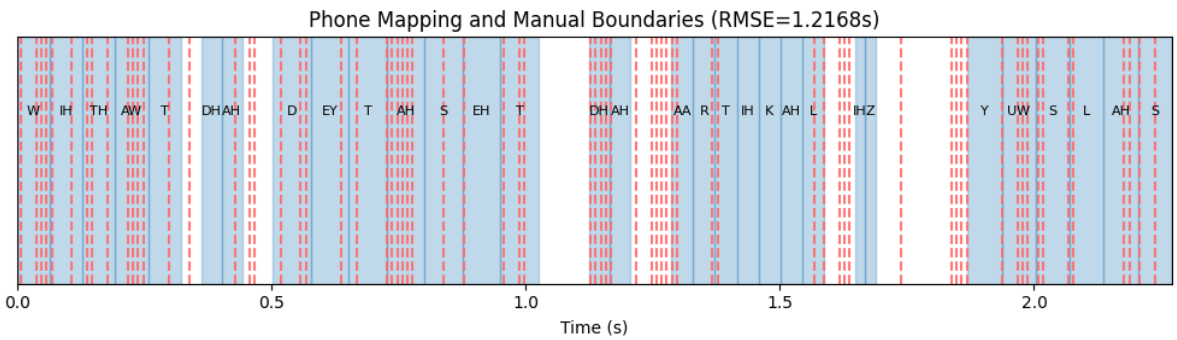
The final stage used a locally cached Wav2Vec2 model to obtain CTC-based alignment information. Character-aligned outputs were converted into approximate phone-level segments using a small lexicon. The manual cepstral boundaries were then compared against the model-derived boundaries using RMSE.

Table 2. RMSE Between Manual Boundaries and Model Alignment

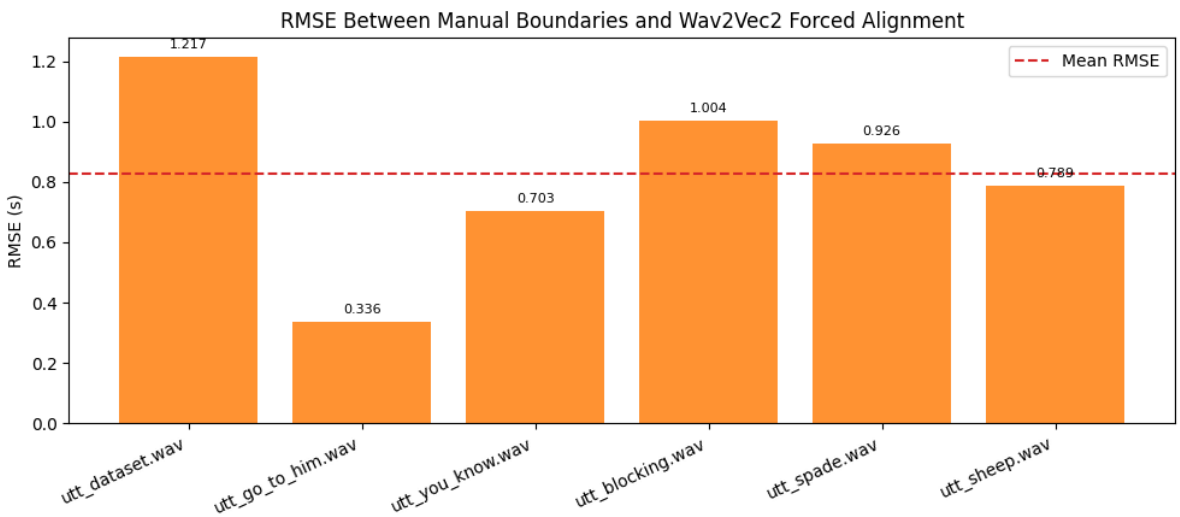
Audio File	RMSE (seconds)
utt_dataset.wav	1.2168

The RMSE result shows that the manual boundary detector captures broad segment changes, but it is still coarser than model-based alignment. This is expected, since voiced/unvoiced detection provides approximate transitions rather than exact phoneme boundaries.

Representative figure:



“without the dataset the article is useless”



6. Discussion

The assignment demonstrates that a full cepstral front-end can be implemented manually using standard signal-processing operations. The MFCC engine reproduced the expected stages of classical speech analysis, while the leakage experiment highlighted the importance of window selection. The voiced/unvoiced detector showed that cepstral structure can be used for segmentation, and the phonetic mapping stage connected the handcrafted pipeline with a modern pretrained Transformer model.

A limitation of this work is that the alignment step used a lightweight proxy rather than a full phone-level forced aligner. Similarly, voiced/unvoiced segmentation is only a coarse approximation of phonetic boundaries. Even so, the system is sufficient to demonstrate the relationship between cepstral analysis, windowing, periodicity, and linguistic segmentation.

7. Conclusion

In conclusion, this assignment successfully implemented a manual speech-processing pipeline for cepstral feature extraction, spectral leakage analysis, voiced/unvoiced detection, and phonetic comparison. The results show that cepstral analysis remains a useful and interpretable approach for studying speech structure. The final RMSE of 1.2168 seconds reflects approximate agreement between manual boundary detection and model-based alignment.